

Sort Color (Dutch National Flag)

→ Brute force

Code:-

```
int[] arr = new int[n];
int i, j = 0;
for (i = 0 to n)
    if (i == 0)
        arr[j++] = 0;

for (i = 0 to n)
    if (i == 1)
        arr[j++] = 1;

for (i = 0 to n)
    if (i == 2)
        arr[j++] = 2;
```

Problem

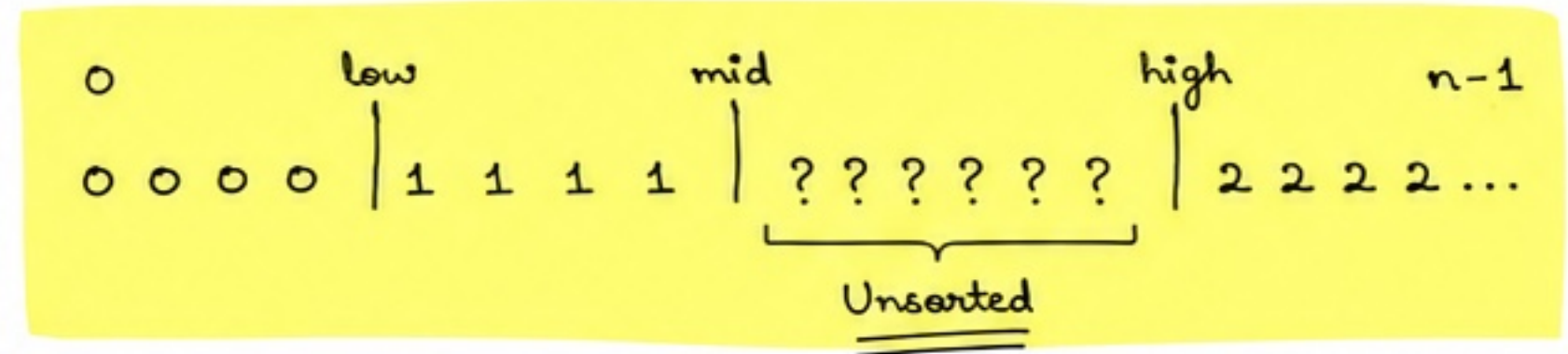
S.C → $O(n)$;
because we use
extra array
to sort the
array.

→ Optimize (Dutch National Flag)

Code:-

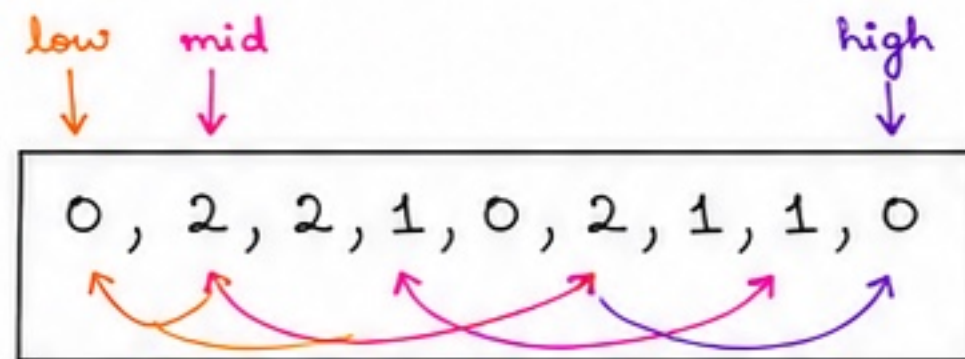
```
low = 0;
mid = 0;
high = n - 1
```

```
while (mid ≤ high) {
    if (arr[mid] == 0) {
        swap(low, mid);
        low++;
        mid++;
    }
    else if (arr[mid] == 2) {
        swap(high, mid);
        high--;
    }
    else {
        mid++;
    }
}
```



Dry Run:-

arr = [0, 2, 2, 1, 0, 2, 1, 1, 0]



→ [0, 0, 0, 1, 1, 2, 2, 2, 2]

low mid high

↓ ↓ ↓

0 n n-1

- ① (mid = 0) → 0 → else
- ② (mid = 1) → 2 → swap(high, mid)
- ③ (mid = 1) → 2 → swap(high, mid)
- ④ (mid = 1) → 1 → else
- ⑤ (mid = 2) → 2 → swap(high, mid)
- ⑥ (mid = 2) → 1 → else
- ⑦ (mid = 3) → 1 → else
- ⑧ (mid = 4) → 0 → swap(low, mid)
- ⑨ (mid = 5) → 2 → swap(high, mid)
- ⑩ (mid = 5) → 1 → else